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THE SUPERCONDUCTING TRANSITION TEMPERATURE FOR VERY LARGE ELECTRON-PHONON COUPLING

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A rigorous derivation of the range of validity of the asymptotic relation $k_B T_c \approx 0.1827[\lambda\langle\omega^2\rangle]^{\frac{1}{2}}$ is presented.

In a paper extending McMillan's classic work[1] on the superconducting transition temperature (T_c), Allen and Dynes[2] found a new result: for "sufficiently large" coupling constant λ, T_c behaves asymptotically as $\lambda^{\frac{1}{2}}$, rather than saturating at a value independent of λ as in earlier[1] theories. The criterion for λ being 'sufficiently large' was too loosely investigated in ref. 2, and is wrong in extreme instances.[3] Further studies[4,5] of the asymptotic regime have now appeared, clarifying the situation. This note presents a simple modification of an important result of ref. 2 which eliminates the error pointed out in ref. 3. For simplicity, we consider explicitly only the case in which the Coulomb pseudopotential, μ^*, is zero.

Our new result for the asymptotic limit of T_c is

$$\text{Limit}_{\lambda\langle\omega^2\rangle >> \omega_{max}^2} \quad k_B T_c = 0.1827[\lambda\langle\omega^2\rangle]^{\frac{1}{2}} , \qquad (1)$$

where ω_{max} is the maximum phonon frequency and

$$\lambda\langle\omega^2\rangle \equiv 2\int_0^\infty d\omega\ \alpha^2F(\omega)\omega , \qquad (2)$$

with

$$\lambda \equiv 2\int_0^\infty d\omega\ \frac{\alpha^2F(\omega)}{\omega} . \qquad (3)$$

Equation (1) and the corresponding result of ref. 2 differ only in that $\lambda\langle\omega^2\rangle >> \omega_{max}^2$ in the former is replaced by $\lambda >> 1$ in the latter with the further statement that $\lambda \gtrsim 10$ is sufficient for accuracy to within a few percent. For spectral functions α^2F of 'ordinary' shape (where $\langle\omega^2\rangle$ is no smaller than $\omega_{max}^2/10$) this is quite consistent with (1) and thus essentially correct. However, for highly dispersive spectra it is possible to satisfy $\lambda >> 1$ and yet lie well outside the regime $\lambda\langle\omega^2\rangle >> \omega_{max}^2$ where (1) is relevant. The condition $\lambda >> 1$ is thus necessary but not sufficient. We now show that the condition $\lambda\langle\omega^2\rangle >> \omega_{max}^2$ is sufficient.

Using the Matsubara imaginary frequency version[6,7] of the Eliashberg theory[8] for T_c, the spectral function only appears in integrals of the type

$$\lambda(n) \equiv 2\int_0^{\omega_{max}} \frac{d\omega\ \alpha^2F(\omega)\ \omega}{\omega^2+(2\pi n k_B T_c)^2} . \qquad (4)$$

Further, only integers $n>0$ need be considered.[2] Hence, for large T_c the term ω^2 can be neglected relative to $(2\pi n k_B T_c)^2$ in the denominator of (4), giving

$$\lambda(n) \approx \lambda\langle\omega^2\rangle/(2\pi n k_B T_c)^2 \quad + \ldots . \quad (n>0). \qquad (5)$$

If T_c is not large enough correction terms[5] involving frequency moments ($\langle\omega^4\rangle$, etc.) become important in (5). Thus a necessary condition for T_c to depend only on $\lambda\langle\omega^2\rangle$ is that T_c be large enough that only the first term of (5) is important. Clearly this is true if

$$2\pi k_B T_c >> \omega_{max} . \qquad (6)$$

Condition (6) is also a sufficient condition for the asymptotic limit (1) to apply, because T_c, λ and $\langle\omega^2\rangle$ then enter the theory only in the combination $\lambda\langle\omega^2\rangle/T_c^2$. Therefore, the T_c

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equation in this limit must take the form $T_c/\sqrt{\lambda\langle\omega^2\rangle} = c(\mu^*)$ where the function c depends only on μ^*. Numerical solution of the gap equations gives $c(0) = 0.1827$.

Condition (6) is not a very satisfactory criterion for using the asymptotic formula (1), because it refers explicitly to T_c rather than λ. It leaves open the objection that something else in the Eliashberg theory might prevent (6) from being attained, making the limit (1) irrelevant or even wrong. Therefore we seek to express condition (6) as a condition on the spectral function α^2F. This can be done using the following inequalities:

$$\frac{\lambda\langle\omega^2\rangle}{\omega_{max}^2+(2\pi k_B T_c)^2} \leq 2 \int_0^{\omega_{max}} \frac{d\omega\ \alpha^2F(\omega)\omega}{\omega^2+(2\pi k_B T_c)^2} \leq 1 . \quad (7)$$

The left inequality is obvious; the right one is proved in ref. 2. A consequence of (7) is

$$\lambda \frac{\langle\omega^2\rangle}{\omega_{max}^2} \leq 1 + \left(\frac{2\pi k_B T_c}{\omega_{max}}\right)^2 . \quad (8)$$

Thus the condition

$$\lambda\langle\omega^2\rangle >> \omega_{max}^2 \quad (9)$$

is sufficient to guarantee (6), i.e. sufficient to assure the correctness of equation (1). Since there is nothing in Eliashberg theory to prevent (9), the asymptotic regime is <u>in principle</u> physically accessible, although in practice not yet attained.

It is interesting to note that equation (1) can be written compactly in terms of the first moment of α^2F:

$$\text{Limit}_{x >> \omega_{max}} \quad k_B T_c = 0.2584\, x , \quad (10)$$

where

$$x \equiv \left[\int_0^{\omega_{max}} d\omega\ \alpha^2F(\omega)\omega\right]^{\frac{1}{2}} . \quad (11)$$

(For elements $x^2 = \eta/2M$ where η is the McMillan-Hopfield parameter[1,9] and M is the ionic mass.) This suggests that in the limit of very strong electron-phonon coupling it is the first moment of α^2F, rather than λ, which is the single most important parameter characterizing α^2F.[2,10,11,12]

Finally, we present numerical calculations which illustrate how large λ must be in order for eq. (1) to be accurate to within 5%. These are shown in table I for power law spectra of the form $\alpha^2F(\omega) = (n\lambda/2)(\omega/\omega_{max})^n\theta(\omega_{max}-\omega)$ with various choices of n. Note that the critical value of $\lambda\langle\omega^2\rangle/\omega_{max}^2$ shows less variation than does that of λ. This is particularly evident in the unphysical limit $n \to 0$ (keeping $n\lambda$ finite) where λ diverges. In this limit it is clear that λ is not a very useful parameter while the first moment of α^2F is.[12]

TABLE I

For power-law spectral functions ($\alpha^2F \propto \omega^n$), $\lambda_{(5\%)}$ is the numerical value which λ muxt exceed for eq. (1) to be accurate to within 5% (the other quantities are similarly defined). The exponent $n = \infty$ corresponds to an Einstein spectrum, and the (unphysical) exponent $n = o$ is discussed in the text. To achieve accuracy to within 2%, λ must be larger than $\lambda_{(5\%)}$ by a factor of ~ 2.5.

n	$\lambda_{(5\%)}$	$(\langle\omega^2\rangle\lambda/\omega_{max}^2)_{(5\%)}$	$(x/\omega_{max})_{(5\%)}$
0	∞	3.4	1.3
½	18.8	3.8	1.4
1	12.3	4.1	1.4
2	9.2	4.6	1.5
∞	7.0	7.0	1.9

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